JUSTIFICATION FOR EQUIPMENT

Project Title : ***Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay***

Principal Investigator : For. Divina P. Chica
Implementing Institution (HEI) : Partido State University-Tinambac Campus

Period Covered : Two (2) years
Total Project Cost : Php 4,500,000.00

No.	Equipment	Justification	Total Price (Php)
1	GPS units Php 25,000.00/ item X 1 item Specifications/ features: • Display Resolution:200x265 pixels • Battery Type: Lithium Ion • Water Resistance Level: IPX-7 • Additional Features: Sensors and TOPO Mapping • Weight: 7.5 oz	GPS units are essential in accurately locating sample plots and specific communities. Using GPS, it will be easier to mark the locations of vegetation, which will be used for map generation. Without GPS, it will be difficult to accurately plot the sampling plots.	25,000.00
2	Hypsometer Php 23,000.00/ item X 3 items Specifications/ features: • Measurement Range: 8 - 1,750 yd. • Increment Reading: 0.3 yds • Magnification: 6x • Angular Field of View (Real): 7.5° • Viewfinder Display: meters/yards • Exit Pupil: 3.5mm Eye Relief: 18.0mm Diopter Adjustment: +/- 4° • Power Source: 1 CR2 Lithium Battery • Size (Length x Width x Height): 4.3 in. x 2.9 in. x 1.7 in. • Weight (Without Batteries): 6.0 oz • Objective Diameter: 21mm • With Angle Compensation • Waterproof	Hypsometers are essential in accurate estimation of tree height and distance. The procurement of three (3) hypsometers will facilitate efficient data collection. This will aid the researchers in producing quality research, especially when it comes to vegetation and carbon stock assessment in the mangrove forest of San Miguel Bay. If this equipment is not purchased, field activities will heavily rely on traditional survey methods that are prone to significant errors.	75,000.00
3	STATA Software License (1 year period: 2nd year of project implementation) Php 42,000.00/ item X 1 item Specifications/ features: • Multiple users (concurrent use) • STATA/SE Annual Up to 32, 676 valiables 20 seconds runtime for logistic regression	STATA is a powerful tool that is widely used by researchers all over the world to produce sound research results., especially in econometrics and regression analysis that are needed in this study. Without this software, it will be difficult to handle complicated data that requires complicated statistical analysis.	42,000.00
4	Unmanned Aerial Vehicle/Drone Php 96,000.00/ unit x 1 unit Specifications/ features: • 1/2.3" CMOS; 12 mp sensor • 24 mm at f/2.8 focal & aperture • 430 g in weight • 168x83x49 mm folded size / 168x184x64 mm unfolded size	Drone is a useful tool for remote sensing and thus, have many applications in the component of mangrove assessment of San Miguel Bay. • Capture real-time aerial view of mangrove stand to be use as basis for crafting sampling plan. • Capture aerial shots to provide us qualitative status of the stands	96,000.00

	• With GPS & GLONASS • With downward, and anti-collision back and front sensors • 16 minutes effective battery time • Wi-Fi: 2.4 GHz & 5.8 GHz • 3-axis mechanical stabilization • Multi video modes • 100 mbps max bitrate • Maximum speed of 28.8 kph (normal mode) and 68.4 kph (sport mode)	• Survey the mangrove stands pre- and post-sampling • Survey and reach highly-isolated mangrove stands of SMB • Capture aerial shots of mangrove strands to be used in photo-documentation and manual making. This tool will be highly effective in crafting the sampling plan of mangrove stands since the proponents are dealing with large quantities of mangrove area.	
5	Tablet Php 54,000.00/ unit x 1 unit Specifications/ features: • 10.9 inch liquid retina IPS LCD with scratch resistant glass • Octa-core, Apple M1 chipset, iPadOS 15.4 • 8GB RAM 64GB • 12 mp, f/1.8 with HDR and video: 1080@30.60fps • 5G enabled network and Nano-Sim slot • With WiFi 802.11;With Bluetooth 5.0; GPS; USB Type-C 3.1 • With fingerprint sensor, accelerometer, gyro, compass, and barometer • Li-Ion (28.6 Wh) non removable battery • Dimensions: 247.6 x 178.5 x 6.1 mm • Weight: 462 g	The tablet will be used as the main operating unit for the drone. The high operating system of the tablet is needed to ensure the efficient use and functioning of the drone and its captured data. The choice of buying tablet with sim card slot is to cater one time password (OTP) for verification of signing into accounts in apps for operating a drone.	54,000.00
6	Digital Single-lens Reflex (DSLR) Camera Php 43,000.00 Specifications/ features: • 22.3 mm x 14.9 mm CMOS • Approx. 24.10 mp as effective pixels; 25.80 mp as total pixels • 3:2 aspect ratio • EOS integrated cleaning system • DIGIC 8 image processor • EF and EF-S lens mount • 1.6x focal length of lens • With optical image stabilizer • With optical viewfinder: 0.87x magnification; fixed focusing screen • With LiveView on LCD screen • LCD screen: vari-angle touchscreen 7.7 cm ; 170 degrees viewing angle • With built-in flash: ISO 100 9.8 meters; flash coverage up to 18 mm focal length; Auto. Manual flash • Hi-Speed USB with Wi-Fi (2.4 Ghz) and Bluetooth (BLE) • Storage: SD, SDHC, SDXC (UHS Speed Class 1) • Dimensions: 122.4 x 92.6 x 69.8 mm • Weight: 450 g	The camera will be used to capture high-quality images that will be used for the photo-documentation of the research activities. So as, capture images of the mangrove stands of San Miguel Bay to be used in making the manual as one of the proposal's output.	43,000.00
TOTAL			**Php 335,000.00**

SUBMITTED BY:

For. DIVINA P. CHICA
Principal Investigator (Project Leader)
Partido State University-Tinambac Campus

REVIEWED AND ENDORSED BY:

ROMEO B. MORCILLA, Ph.D.
Campus Dean
Partido State University-Tinambac Campus

RAUL G. BRADECINA, Ph.D.
University President
Partido State University
Goa, Camarines Sur